

How ARCHITECTURE Is Becoming INTELLIGENT



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In her book *The Fountainhead*, Ayn Rand portrays the protagonist, Howard Roark, as an individualistic young architect who chooses to struggle in unimportance rather than compromise his artistic and personal vision. The book follows his fight to practice what the public sees as modern architecture, which he considers to be superior, despite an establishment focused on tradition-worship.

In the book *Architecture's New Media: Principles, Theories, and Methods of Computer-Aided Design*, authors Yehuda E. Kalay and William J. Mitchell talk about computer-aided architectural design (CAAD) and its ability to model and manipulate objects, beyond just graphical representation. They also go into how CAAD can be used to forecast the performance of design solutions, create new design solutions via algorithmic and alternate methods and organise huge amounts of information.

Software programming, knowledge of complex algorithms, genetic engineering, artificial intelligence, cloning and nanotechnology are the future areas for architects. Perhaps, there are many Howard Roark's today who are struggling to get CAAD accepted by the

public who still prefers to stand by the conventional approach.

Years ago when computer-aided design (CAD) was nonexistent, architects evolved their plans while drawing and they did their thinking along the way. The advent of CAD demanded them to make accurate decisions at each step and input different dimensions into the system. However, this in a way became a sort of barrier to creativity. The study of information technology (IT) applications in construction is relatively a young field of research, still nascent among the large family of academic disciplines and thus devoid of a solid methodological foundation.

During the early years, IT in construction and architecture was almost entirely used to support activities that could be labelled as creation of new information. Analysis programs for structural analysis and other such applications depended on tiresome manual preparation of input data (for example, in the form of punched cards). During the early 1980s, use of CAD started to spread; however, focus was more on support for creation and viewing of data. For example, many people were able to sit by a screen and see the same image during a design session.

Emphasis was also on image retrieval on multiple terminals connected to the same dedicated super-mini on which CAD software was running. Support for making information available came in the form of very costly A1/A0 plotters, which enabled the plotting of drawings that matched manually-produced drawings and were sent to copying services before actual distribution.

Efficiency of drawing production (especially in connection with changes) increased at the most by a factor of two to three. Today, the situation has changed radically. Developments in LAN and WAN networks, the Internet, mobile phones, video-conferencing and so on has extend-

Fig. 1: Harbin Opera House in Harbin, North China (Image courtesy: www.huffonandcrow.com)

